

Children and Their Parents' Labor Supply: Evidence from Exogenous Variation in Family Size (Angrist and Evans, 1998) Paper Review & Analysis

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1 Research Question

This section presents the central research questions and the main motivation behind the study by Angrist and Evans.

1.1 Main Research Question

The paper studies whether having an additional child affects parents' labor supply. More specifically, it focuses on the transition from two to three or more children and asks whether this fertility increase changes the probability of working, annual weeks worked, weekly hours worked, and labor income. A central question is whether the effect differs between mothers and fathers. While previous literature mainly focused on women, Angrist and Evans also examine whether men adjust their labor supply in response to an additional child. The paper further investigates whether the effect varies by education and husbands' earnings, and whether the labor supply response is temporary or persistent over time.

1.2 Key Motivation and Goals

The main empirical challenge is endogeneity. Fertility is not randomly assigned: families with more children may differ systematically from families with fewer children in preferences, income, labor market attachment, or other unobserved characteristics. For this reason, simple OLS estimates may be biased. To address this issue, Angrist and Evans use instrumental variables based on sibling sex composition and twin births. The key instrument is whether the first two children are of the same sex. Since parents often prefer a mixed sibling-sex composition, families with two boys or two girls are more likely to have a third child. At the same time, the sex of children can be considered plausibly random, making Same Sex a credible source of exogenous variation in family size. The goal of the paper is therefore to estimate the causal effect of fertility on labor supply and to compare OLS and IV estimates in order to understand how much selection affects the observed relationship.

2 Main Empirical Results

This section summarizes the main findings of the paper, focusing on the causal effect of an additional child on parents' labor supply.

2.1 Central Findings

The main result is that having more than two children significantly reduces mothers' labor supply, although the causal effect estimated by IV is smaller than the corresponding OLS estimate. In the 1980 Census sample, OLS estimates suggest that having a third child reduces the probability that a woman worked by 17.6 percentage points, annual weeks worked by almost 9 weeks, weekly hours

by about 6.7 hours, and yearly income by approximately \$3,768. Using the Same Sex instrument, the estimated effects remain negative but become smaller. The IV estimates suggest that having more than two children reduces the probability of working by around 12 percentage points, weekly hours by 4.59 hours, annual weeks worked by 5.66 weeks, and yearly income by about \$1,961. This difference between OLS and IV suggests that women with larger families differ from women who stop at two children in unobserved ways. The 1990 results show a similar pattern. The negative effect on female labor supply remains visible, although the magnitude is somewhat smaller: the probability of working falls by around 9.2 percentage points, weekly hours by 4.08 hours, annual weeks by 5.66 weeks, and earnings by roughly \$2,100.

The consistency of the results across the two Census years strengthens the interpretation that the negative effect is not specific to a single sample or period. At the same time, the smaller magnitude in 1990 suggests that changes in female labor-force participation, childcare availability, or broader labor-market conditions may have reduced the employment cost of additional childbearing over time.

2.2 Heterogeneous Effects

The paper also shows that the effect of an additional child is not homogeneous across households. In particular, the labor supply response varies by women's education and husbands' earnings. For women with more than a high school diploma, the negative effect of an additional child is generally smaller. By contrast, less educated women experience larger reductions in labor supply, especially in the 1990 results. One possible explanation is that more educated women may have better access to flexible jobs, higher wages, or childcare options. The authors also find that the negative effect on wives' labor supply tends to be smaller when husbands have higher earnings. Overall, these results suggest that the labor market cost of an additional child is stronger for more constrained households.

2.3 Fathers' Labor Supply

A key result concerns fathers. While OLS estimates sometimes suggest a positive relationship between family size and male labor supply, this relationship disappears once fertility is instrumented. The IV estimates show no statistically significant causal effect of an additional child on fathers' probability of working, weekly hours, or annual weeks worked. This contrasts sharply with the results for mothers. The evidence therefore suggests that the labor supply adjustment to an additional child is mainly borne by women, while men do not significantly increase or reduce their labor supply in response.

2.4 Duration of the Effect

The paper also studies whether the labor supply effect is temporary or persistent. The evidence suggests that the negative impact of children on mothers' labor supply is strongest when children are young and gradually fades as they grow older. This conclusion is supported by comparing the Same Sex and twins instruments. The Same Sex instrument induces parents to have a third child later, while the twins instrument generates an additional child at the second birth. Since the estimates differ across these instruments, the authors infer that the age of the child matters. Overall, the findings suggest that childbearing does not cause a permanent withdrawal from the labor market. Instead, it leads to a temporary reduction in mothers' labor supply during the years in which childcare needs are more intense. The effect becomes much smaller and approximately negligible when the child reaches adolescence.

3 Paper Replication

This section summarizes the replication of the main empirical results focusing on descriptive statistics, validity of the instruments, Wald estimates, OLS and 2SLS results and heterogeneity analysis.

3.1 Descriptive Statistics

The descriptive statistics replicate the main variables used in the original analysis, using women aged 21–35 with two or more children from the 1980 and 1990 Census PUMS. The instrumental variables behave as expected. The variables indicating whether the first or second child is a boy have means close to 0.51, while the Same Sex variable has a mean close to 0.5. This supports the idea that sibling sex composition is approximately randomly distributed across families. Women's labor force

attachment increased substantially between 1980 and 1990. The share of women working for pay rose from 0.561 to 0.667, annual weeks worked increased from 20.6 to 26.4, and average labor income increased from \$7,622 to \$9,321. Husbands, instead, show much higher and more stable labor force participation, above 0.97, with weekly hours close to 48. This already anticipates one of the main findings: fathers' labor supply is much less responsive to family size than mothers' labor supply.

Variable	1980 Census			1990 Census		
	All Women	Married Wives	Husbands	All Women	Married Wives	Husbands
Children ever born	2.55 (0.81)	2.51 (0.78)		2.50 (0.77)	2.48 (0.75)	
More than 2 children	0.399 (0.490)	0.381 (0.486)		0.369 (0.483)	0.362 (0.481)	
Boy 1st	0.511 (0.500)	0.514 (0.500)		0.513 (0.500)	0.515 (0.500)	
Boy 2nd	0.511 (0.500)	0.513 (0.500)		0.511 (0.500)	0.512 (0.500)	
Two boys	0.263 (0.440)	0.265 (0.441)		0.263 (0.440)	0.264 (0.441)	
Two girls	0.240 (0.427)	0.238 (0.426)		0.238 (0.426)	0.236 (0.425)	
Same sex	0.503 (0.500)	0.503 (0.500)		0.501 (0.500)	0.500 (0.500)	
Twins-2	0.0089 (0.0939)	0.0087 (0.0926)		0.0115 (0.1068)	0.0109 (0.1038)	
Age	30.0 (3.6)	30.2 (3.5)	32.6 (4.4)	30.4 (3.5)	30.6 (3.3)	33.4 (4.8)
Age at first birth	20.9 (3.1)	21.5 (3.0)	24.0 (3.8)	21.7 (3.5)	22.2 (3.5)	25.0 (4.8)
Worked for pay	0.561 (0.496)	0.529 (0.499)	0.977 (0.149)	0.667 (0.471)	0.670 (0.470)	0.969 (0.173)
Weeks worked	20.6 (22.2)	19.0 (21.8)	47.9 (10.5)	26.4 (22.9)	26.5 (22.8)	47.1 (12.0)
Hours/week	18.7 (18.9)	16.7 (18.3)	43.5 (12.2)	22.7 (19.1)	22.3 (18.9)	44.2 (13.4)
Labor income (\$1995)	7622 (11617)	6697 (10977)	41609 (26713)	9321 (12759)	9355 (12892)	35534 (29659)
Family income (\$1995)	45536 (28682)	51064 (27646)		41874 (33971)	47888 (33986)	
Non-wife income (\$1995)		44367 (26282)			38533 (31197)	

Notes: The samples include women aged 21-35 with two or more children except for women whose second child is less than a year old. In the 1980 PUMS, the married women sample refers to women who were married at the time of their first birth, married at the time of the survey, and married once. In the 1990 PUMS, the married women are those married at the time of the Census.

3.2 Fertility and Sex Mix

The first empirical step is to examine whether the sex composition of the first two children affects the probability of having a third child. In the 1980 PUMS, families with one boy and one girl had a 36.9% probability of having a third child. This probability increased to 41.8% for families with two boys and to 43.7% for families with two girls. Overall, families with two children of the same sex had a 42.7% probability of having a third child, compared to 36.9% for mixed-sex families.

Mixed sex families had a 33.8% probability of a third birth, while same-sex families had a probability close to 40%. The difference is also similar among married women, where the same-sex effect ranges from about 6.6 to 6.8 percentage points. Across years and samples, families with two children of the same sex are consistently more likely to have a third child.

This supports the relevance of the Same Sex instrument.

3.3 Validation of the Instrument

The first stage regressions confirm that Same Sex is a strong and statistically significant predictor of having more than two children. In both 1980 and 1990, having two children of the same sex increases the probability of another child by roughly 6 percentage points, with slightly larger effects among married women. The estimates also show some differences between two boys and two girls, with two girls often generating a slightly stronger fertility response. However, these differences are smaller than the overall Same Sex effect. The key result is that sibling sex composition strongly predicts additional fertility, satisfying the relevance condition for the instrument.

This is a crucial step because a weak first stage would make the IV estimates unreliable. In this case, the repeated significance of Same Sex across samples and specifications indicates that the instrument has enough predictive power to be used in the second stage.

Table 3: Fraction of Families That Had Another Child by Parity and Sex of Children

Bottom Panel: Sex of first two children in families with two or more children								
	1980 PUMS		1990 PUMS		1980 PUMS		1990 PUMS	
	All women				Married women			
	Fraction of sample	Fraction that had another child	Fraction of sample	Fraction that had another child	Fraction of sample	Fraction that had another child	Fraction of sample	Fraction that had another child
one boy, one girl	0.497	0.369 (0.001)	0.499	0.338 (0.001)	0.497	0.348 (0.001)	0.500	0.328 (0.001)
two girls	0.240	0.437 (0.002)	0.238	0.405 (0.002)	0.238	0.426 (0.002)	0.236	0.402 (0.002)
two boys	0.263	0.418 (0.002)	0.263	0.396 (0.002)	0.265	0.403 (0.002)	0.264	0.391 (0.002)
(1) one boy, one girl	0.497	0.369 (0.001)	0.499	0.338 (0.001)	0.497	0.348 (0.001)	0.500	0.328 (0.001)
(2) both same sex	0.503	0.427 (0.001)	0.501	0.400 (0.001)	0.503	0.414 (0.001)	0.500	0.396 (0.001)
difference (2) - (1)	—	0.058 (0.002)	—	0.062 (0.002)	—	0.066 (0.002)	—	0.068 (0.002)
Notes: The samples are the same as in Table 2. Standard errors are reported in parentheses								

Notes: The samples are the same as in Table 2. Standard errors are reported in parentheses

Table 6 — OLS Estimates of More than 2 Children Equations

Variable	All women			Married women		
	(1)	(2)	(3)	(4)	(5)	(6)
1980 PUMS						
Boy 1st		-0.0083 (0.0015)	-0.0004 (0.0021)		-0.0123 (0.0018)	-0.0015 (0.0025)
Boy 2nd		-0.0079 (0.0015)			-0.0107 (0.0018)	
Same sex	0.0579 (0.0016)	0.0586 (0.0015)		0.0661 (0.0019)	0.0675 (0.0018)	
Two boys			0.0507 (0.0021)			0.0567 (0.0025)
Two girls			0.0665 (0.0021)			0.0782 (0.0026)
With other covariates	no	yes	yes	no	yes	yes
R ²	0.003	0.096	0.096	0.005	0.090	0.090
1990 PUMS						
Boy 1st		-0.0079 (0.0017)	-0.0081 (0.0023)		-0.0095 (0.0019)	-0.0087 (0.0026)
Boy 2nd		0.0002 (0.0017)			-0.0009 (0.0019)	
Same sex	0.0617 (0.0017)	0.0601 (0.0017)		0.0685 (0.0019)	0.0678 (0.0019)	
Two boys			0.0603 (0.0023)			0.0669 (0.0026)
Two girls			0.0600 (0.0024)			0.0686 (0.0027)
With other covariates	no	yes	yes	no	yes	yes
R ²	0.004	0.081	0.081	0.005	0.081	0.081

Notes: Other covariates in the models are indicators for Age, Age at first birth, Black, Hispanic, and Other race. The variable Boy 2nd is excluded from columns (3) and (6). Standard errors are reported in parentheses.

3.4 Wald Estimates

Variable	1980 PUMS			1990 PUMS			1980 PUMS		
	Wald estimate using as covariate:			Wald estimate using as covariate:			Wald estimate using as covariate:		
	Mean difference by Same sex	More than 2 children	Number of children	Mean difference by Same sex	More than 2 children	Number of children	Mean difference by Twins-2	More than 2 children	Number of children
More than 2 children	0.0579 (0.0016)	—	—	0.0617 (0.0017)	—	—	0.6068 (0.0008)	—	—
Number of children	0.0737 (0.0026)	—	—	0.0816 (0.0028)	—	—	0.8044 (0.0115)	—	—
Worked for pay	-0.0082 (0.0016)	-0.142 (0.027)	-0.111 (0.021)	-0.0057 (0.0017)	-0.092 (0.027)	-0.070 (0.021)	-0.0483 (0.0085)	-0.080 (0.014)	-0.060 (0.011)
Weeks worked	-0.4030 (0.0708)	-6.96 (1.21)	-5.46 (0.95)	-0.3455 (0.0825)	-5.60 (1.32)	-4.24 (1.00)	-2.0699 (0.3706)	-3.41 (0.61)	-2.57 (0.46)
Hours/week	-0.3137 (0.0603)	-5.42 (1.04)	-4.25 (0.82)	-0.2476 (0.0689)	-4.02 (1.11)	-3.04 (0.84)	-2.0254 (0.3140)	-3.34 (0.52)	-2.52 (0.39)
Labor income	-159.0 (37.0)	-2746.9 (635.6)	-2156.0 (499.1)	-133.3 (46.8)	-2161.5 (753.1)	-1634.2 (568.4)	-599.1 (199.5)	-987.2 (328.8)	-744.7 (247.7)
ln(Family income)	-0.0020 (0.0043)	-0.034 (0.075)	-0.027 (0.059)	-0.0108 (0.0055)	-0.175 (0.088)	-0.133 (0.067)	-0.0413 (0.0246)	-0.068 (0.041)	-0.051 (0.031)

Notes: The samples are the same as in Table 2. Standard errors are reported in parentheses.

by 5.42 hours, and yearly income by about \$2,747. In 1990, the estimates are smaller, with a reduction in work probability of around 9.2 percentage points and a reduction in annual weeks worked of about 5.6 weeks. The twins instrument produces smaller estimates. This is consistent with the interpretation that the labor supply effect is stronger when children are younger and fades as they age. Therefore, the comparison between instruments provides evidence on the timing and duration of the fertility effect.

3.5 OLS and 2SLS Estimates

Tables 7 and 8 compare OLS and 2SLS estimates for 1980 and 1990. OLS estimates are negative and large across all female labor supply outcomes, in 1980, it suggests that having more than two children reduces the probability of working by 17.4 percentage points, annual weeks worked by 8.8 weeks, weekly hours by 6.6 hours, and labor income by about \$4,012, while the 2SLS estimates using Same Sex are smaller in absolute value. In 1980, the participation effect falls by 0.138 and the labor income by \$2,631. In 1990, the corresponding estimates are around -0.104 and $-\$2,445$. This confirms that OLS tends to overestimate the causal effect of fertility on female labor supply. For husbands, the 2SLS estimates are generally small and statistically insignificant. This confirms that fathers do not significantly adjust their labor supply in response to an additional child.

3.6 Heterogeneity

The replication also examines whether the effect varies across subgroups. Families are divided according to husbands' income, while women are divided by education level: less than high school, high school graduates, and more than high school. The first stage remains strong across groups, meaning that Same Sex continues to predict the probability of having a third child. In 1980, the first-stage effect ranges from 0.057 to 0.079 across husbands' income groups and from 0.063 to 0.073 across women's education groups. The 2SLS estimates show that the negative effect of an additional child is stronger for poorer and less educated women. By husbands' income, the effect on wives' employment in 1980 is -0.122 for the bottom third, -0.185 for the middle third, and

Variable	All women			Married women			Husbands of married women		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Estimation method	OLS	2SLS	2SLS	OLS	2SLS	2SLS	OLS	2SLS	2SLS
Instrument for <i>More than 2 children</i>	—	Same sex	Two boys, Two girls	—	Same sex	Two boys, Two girls	—	Same sex	Two boys, Two girls
Dependent variable:									
<i>Worked for pay</i>	-0.174 (0.002)	-0.138 (0.026)	-0.131 (0.026) [0.011]	-0.165 (0.002)	-0.128 (0.028)	-0.122 (0.028) [0.088]	-0.006 (0.001)	0.010 (0.009)	0.008 (0.009) [0.015]
<i>Weeks worked</i>	-8.807 (0.071)	-6.703 (1.162)	-6.383 (1.155) [0.013]	-7.972 (0.087)	-6.419 (1.229)	-6.258 (1.219) [0.312]	-0.677 (0.045)	0.626 (0.605)	0.464 (0.599) [0.038]
<i>Hours/week</i>	-6.576 (0.062)	-5.263 (0.996)	-5.000 (0.990) [0.018]	-5.985 (0.074)	-5.126 (1.036)	-4.927 (1.027) [0.138]	0.336 (0.053)	1.071 (0.707)	0.993 (0.703) [0.393]
<i>Labor income</i>	-4012 (37)	-2631 (613)	-2536 (609) [0.167]	-3405 (44)	-2101 (623)	-2126 (618) [0.750]	-892 (110)	-1678 (1488)	-1695 (1478) [0.930]
<i>ln(Family income)</i>	-0.131 (0.005)	-0.033 (0.072)	-0.041 (0.071) [0.316]	-0.133 (0.005)	-0.033 (0.061)	-0.038 (0.060) [0.512]	—	—	—
<i>ln(Non-wife income)</i>	—	—	—	-0.053 (0.005)	0.084 (0.073)	0.068 (0.072) [0.100]	—	—	—

Notes: The table reports estimates of the coefficient on the *More than 2 children* variable in equations (4) and (6) in the text. Other covariates in the models are *Age*, *Age at first birth*, plus indicators for *Boy 1st*, *Boy 2nd*, *Black*, *Hispanic*, and *Other race*. The variable *Boy 2nd* is excluded from equation (6). The *p*-value for the test of overidentifying restrictions associated with equation (6) is shown in brackets. Standard errors are reported in parentheses.

Variable	All women			Married women			Husbands of married women		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Estimation method	OLS	2SLS	2SLS	OLS	2SLS	2SLS	OLS	2SLS	2SLS
Instrument for <i>More than 2 children</i>	—	Same sex	Two boys, Two girls	—	Same sex	Two boys, Two girls	—	Same sex	Two boys, Two girls
Dependent variable:									
<i>Worked for pay</i>	-0.155 (0.002)	-0.104 (0.028)	-0.105 (0.028) [0.799]	-0.147 (0.002)	-0.112 (0.028)	-0.112 (0.028) [0.573]	-0.010 (0.001)	0.011 (0.011)	0.011 (0.011) [0.898]
<i>Weeks worked</i>	-8.725 (0.086)	-6.469 (1.331)	-6.439 (1.330) [0.372]	-8.271 (0.096)	-6.326 (1.323)	-6.319 (1.322) [0.709]	-0.889 (0.054)	0.544 (0.715)	0.551 (0.715) [0.505]
<i>Hours/week</i>	-6.802 (0.073)	-4.635 (1.123)	-4.654 (1.122) [0.500]	-6.384 (0.080)	-4.373 (1.098)	-4.381 (1.098) [0.625]	0.038 (0.059)	0.642 (0.783)	0.660 (0.782) [0.110]
<i>Labor income</i>	-3961 (47)	-2445 (758)	-2442 (758) [0.890]	-3736 (53)	-2789 (765)	-2789 (764) [0.987]	1306 (127)	1376 (1687)	1392 (1687) [0.510]
<i>ln(Family income)</i>	-0.124 (0.006)	-0.151 (0.085)	-0.149 (0.085) [0.267]	-0.106 (0.005)	-0.072 (0.061)	-0.072 (0.061) [0.958]	—	—	—
<i>ln(Non-wife income)</i>	—	—	—	-0.005 (0.006)	0.011 (0.080)	0.010 (0.080) [0.595]	—	—	—

Notes: The table reports estimates of the coefficient on the *More than 2 children* variable in equations (4) and (6) in the text estimated with 1990 Census data. Other covariates in the models are *Age*, *Age at first birth*, plus indicators for *Boy 1st*, *Boy 2nd*, *Black*, *Hispanic*, and *Other race*. The variable *Boy 2nd* is excluded from equation (6). The *p*-value for the test of overidentifying restrictions associated with equation (6) is shown in brackets. Standard errors are reported in parentheses.

−0.078 for the top third. By women’s education, the effect is −0.121 for women with less than high school, −0.147 for high school graduates, and −0.082 for women with more than high school. These findings confirm that the labor supply effect is heterogeneous and that the cost of fertility is larger for more constrained households.

3.7 Final Summary

The replication confirms the main conclusions of Angrist and Evans. Additional childbearing reduces mothers’ labor supply, while fathers’ labor supply remains largely unchanged. The negative effects are stronger for poorer and less educated women, suggesting that the labor market cost of fertility is mainly borne by disadvantaged mothers.

4 Extension

This section proposes two possible extensions of the original Angrist and Evans analysis. The first extension focuses on the functional form of the empirical model, comparing the original linear probability model with an IV-Probit specification. The second extension focuses on the interpretation of the IV estimates, especially their external validity and their Local Average Treatment Effect interpretation.

4.1 First Approach: Robustness Check Using IV-Probit

4.1.1 Motivation: A Possible Structural Weakness

Angrist and Evans estimate their second-stage equations using a linear probability model (LPM). This choice has important advantages. The LPM offers a straightforward interpretation of coefficients and, more importantly, preserves the Local Average Treatment Effect (LATE) interpretation in instrumental variable settings. However, the LPM also has a potential weakness: it does not impose any restriction on predicted probabilities. As a result, fitted values can theoretically fall outside the $[0, 1]$ interval. Since one of the main dependent variables is whether the woman worked for pay, this may represent a possible functional-form concern. For this reason, an individual robustness check is proposed based on an IV-Probit model estimated through a control function approach. This should be interpreted as a robustness exercise, not as a correction of the original paper. Angrist has argued extensively that the LPM is often preferable in IV contexts because nonlinear models make the causal interpretation less transparent. Therefore, the goal is simply to test whether the substantive results change when using a nonlinear probability model.

4.1.2 Control Function Approach

The IV-Probit model is implemented using a control function approach. First, the fertility variable, *more_than_2*, is regressed on the instrument, *SameSex*, and on the relevant control variables using OLS. From this first-stage regression, the residuals, denoted as $\hat{v}$, are recovered. These residuals are then included as an additional explanatory variable in the Probit second stage, where the dependent variable is whether the woman worked for pay. The role of $\hat{v}$ is to capture the part of fertility that may still be correlated with unobserved determinants of female labor supply. The coefficient on $\hat{v}$ is therefore central for the interpretation of the model. If this coefficient is statistically significant, it would provide evidence of endogeneity of the fertility variable within the control function framework. If it is not statistically significant, the model fails to reject exogeneity in this specific nonlinear specification.

4.1.3 Endogeneity Test and Bootstrap Inference

In the empirical results, the coefficient on $\hat{v}$ is not statistically significant, with a p-value of 0.165. Therefore, within this specification, the test fails to reject the null hypothesis of exogeneity. This result should be interpreted carefully. It does not prove that fertility is truly exogenous. Rather, it means that, under the assumptions of the IV-Probit control function model, the data do not provide strong statistical evidence of endogeneity. Limited statistical power may also contribute to this result. An issue is that the control function approach involves a generated regressor, since $\hat{v}$ is estimated in the first stage and then used in the second stage. To account for this generated regressor problem, a bootstrap procedure is implemented. In each bootstrap replication, the data are resampled and both stages are re-estimated. The bootstrap standard errors are very close to the asymptotic ones. In particular, the bootstrap standard errors are 0.072 for *more_than_2* and 0.073 for $\hat{v}$, compared with asymptotic standard errors of approximately 0.070. This suggests that asymptotic inference performs well in this large sample. Moreover, the coefficient on $\hat{v}$ remains statistically insignificant under bootstrap inference, reinforcing the conclusion that the model fails to reject exogeneity within this framework.

4.1.4 Interpretation of IV-Probit Results

A key limitation of the IV-Probit model concerns interpretation. In the original 2SLS framework, the coefficient identifies a Local Average Treatment Effect for compliers, namely families whose fertility decision is affected by the sex composition of the first two children. By contrast, the IV-Probit model does not recover a clean LATE. Instead, it relies on stronger distributional assumptions and produces a model-dependent average effect. This makes the causal interpretation less transparent than in the linear IV model. Moreover, Probit coefficients are not directly interpretable as probability changes. Average Marginal Effects (AMEs) are thus computed. The AME for *more_than_2* is -0.136 , meaning that having more than two children reduces the probability that a woman works by approximately 13.6 percentage points. This estimate is close to the corresponding 2SLS estimate reported by Angrist and Evans, approximately -0.12 for labor-force participation. Therefore, even though the IV-Probit model relies on stronger assumptions, it produces a similar result.

TABLE 1 — SECOND STAGE: IV-PROBIT RESULTS

Variable	Coefficient	Std. Error	z-value	p-value
(Intercept)	-0.094	0.033	-2.848	0.004
More than 2 children	-0.361	0.070	-5.180	0.000
Residuals (v-hat)	-0.097	0.070	-1.387	0.165
Boy 1st	-0.001	0.004	-0.349	0.727
Boy 2nd	-0.010	0.004	-2.530	0.011
Age	0.063	0.002	26.838	0.000
Age at first birth	-0.072	0.003	-21.757	0.000
Black	0.262	0.008	31.052	0.000
Hispanic	-0.119	0.012	-10.191	0.000
Other race	0.060	0.010	5.811	0.000

Notes: Probit model estimated via control-function approach. Dependent variable: *Worked for pay*. Coefficients are Probit index coefficients, not marginal effects. *v_hat* tests for endogeneity within the control-function framework.

TABLE 2 — AVERAGE MARGINAL EFFECTS: IV-PROBIT

Variable	Average Marginal Effect	Std. Error	z-value	p-value
More than 2 children	-0.136	0.026	-5.181	0.000
Residuals (v-hat)	-0.036	0.026	-1.387	0.165
Boy 1st	-0.001	0.002	-0.349	0.727
Boy 2nd	-0.004	0.002	-2.530	0.011
Age	0.024	0.001	26.912	0.000
Age at first birth	-0.027	0.001	-21.796	0.000
Black	0.099	0.003	31.165	0.000
Hispanic	-0.045	0.004	-10.195	0.000
Other race	0.023	0.004	5.812	0.000

Notes: Average marginal effects from IV-Probit control-function model. Dependent variable: *Worked for pay*.

4.1.5 Conclusion of the First Approach

Overall, this robustness check suggests that the main conclusions of Angrist and Evans are not driven by the use of a linear probability model. Although the LPM has the drawback of allowing predicted probabilities outside the $[0, 1]$ interval, it remains attractive because of its simple interpretation and its clear LATE framework. The IV-Probit control function approach provides a useful comparison. It introduces additional assumptions and sacrifices the transparent LATE interpretation, but it yields a very similar estimated effect. This reinforces the central finding of the paper: additional childbearing causally reduces mothers' labor supply.

4.2 Second Approach: External Validity and the LATE Interpretation

A second extension concerns the external validity of the Angrist and Evans identification strategy. The instrumental variable approach identifies a Local Average Treatment Effect (LATE), rather than a population-wide average effect. In this context, the SameSex instrument exploits parental preferences for a mixed sibling-sex composition. As a result, the estimated effect applies only to families whose fertility decisions are influenced by the sex composition of their first two children, i.e. the compliers. This implies that the estimated effect does not necessarily generalize to all families. Compliers may differ systematically from other households in terms of preferences, desired family size, labor-market attachment, or socioeconomic characteristics. For instance, families whose fertility decisions are driven by economic constraints, cultural norms, or career considerations may respond differently to the birth of an additional child. This highlights a standard trade-off in the LATE framework. The instrument provides strong internal validity, since child sex composition is plausibly exogenous. However, the resulting estimate is inherently local and may not correspond to the average treatment effect for the entire population. Therefore, the results should be interpreted as strong causal evidence for families whose fertility is affected by the instrument, rather than as a universal estimate of the effect of childbearing on female labor supply.